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Fangyuan Wang



I am a final-year Ph.D. candidate in Mechanical Engineering at The Hong Kong Polytechnic University. My research focuses on developing intelligent systems for long-horizon robotic manipulation, with particular expertise in vision-language-action models, subgoal planning, and embodied AI for mobile manipulation tasks.

RESEARCH EXPERIENCE

Human-like Behavior Generation for Humanoid Robots

Sep 2024 — Present

Supervised by Prof. David Navarro-Alarcon.

Developed a two-stage cross-embodiment framework for humanoid robots that generates context-aware, end-effector-constrained human-like poses, and retargets motions to diverse humanoid morphologies through shape-adaptive body control.

Long-Horizon Manipulation for Robotic Tasks

Sep 2022 — Present

Supervised by Prof. David Navarro-Alarcon.

- Designed a two-stage **Vision-Language-Action** pipeline by first predicting proprioceptive subgoal representations, then generating precise joint-space actions for complex manipulation tasks.
- Introduced **Instruction-Augmented Long-Horizon Planning (IALP)**—a closed-loop system that grounds natural-language commands into PDDL via multimodal perception and LLM reasoning.
- Developed **Explicit-Implicit Subgoal Planning (EISP)**—a hybrid framework coupling explicit-implicit subgoal generation to produce feasible, high-value subgoal sequences, enabling robots to solve long-horizon tasks more efficiently.

Byzantine Defense in Decentralized Learning

Oct 2021 — Sep 2022

Supervised by Prof. Jian Hou.

Designed a credibility-assessment aggregation algorithm that evaluates node reliability through long-term behavioral analysis, achieving 95% accuracy in Byzantine-resilient convergence while maintaining robustness against up to 40% adversarial nodes.

Active Flow Control and Optimal Sensor Placement

Aug 2021 — Sep 2022

Supervised by Prof. Ning Gui during visiting research at Central South University.

Optimized sensor placement for 2D Kármán vortex street drag reduction using distributed RL. Achieved 25% drag reduction by identifying optimal 5-sensor configuration from 151 candidates using attention-based PPO with A3C multi-environment architecture.

Multi-Agent Resilient Consensus

Jul 2020 — Jun 2021

Supervised by Prof. Jian Hou.

Bridged multi-agent RL with resilient consensus by developing adaptive adjacency weight learning via DDPG to identify and reject adversarial neighbors. Proposed Q-consensus algorithm to eliminate neural network dependency while maintaining convergence guarantees.

SELECTED PUBLICATIONS

Published Journal Articles

